

POKHARA UNIVERSITY

Level: Bachelor
 Programme: BE
 Course: Electronic Devices and Circuits (New)

Semester: Fall

Year : 2023
 Full Marks: 100
 Pass Marks: 45
 Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

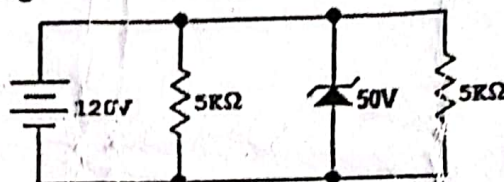
The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Fermi-level in an intrinsic semiconductor. Also, prove that "The Fermi level lies in mid-way between the conduction and valance bands." 7

- b) Determine the following parameter for a given circuit diagram. 8

- i: The output voltage
- ii. Voltage drop across the series resistance
- iii. Current through the Zener diode

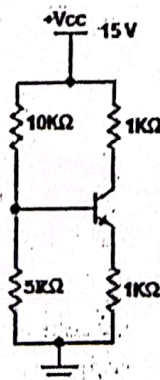


2. a) Draw the input and output characteristics curve for a common emitter configuration. Also, explain why CE configuration is used for amplification. 7

OR

Explain the working principle of NPN transistor with necessary circuit diagram. Show that $I_c = \beta I_B + (1 + \beta) I_{CBO}$.

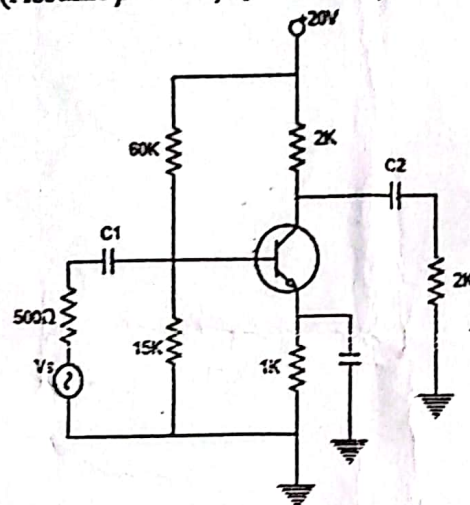
- b) Determine the operating point and locate it in the load line for a given circuit below Assume $\beta = 100$. 8



3. a) What is Barkhausen Criterion for oscillation? Design a RC phase shift oscillator to operate at a frequency of 8 kHz. 8

OR

- Explain Barkhausen criteria of oscillation? Design a Wein-bridge oscillator to operate at a frequency of 24 kHz.
- b) Explain the working principle of monostable multivibrator using an operational amplifier with necessary circuit diagram and waveform. 7
4. a) Compare between full-wave and half-wave rectifier. Explain the working principle of full-wave center tap rectifier. 7
- b) Explain the construction and working principle of MOSFET. Point out the difference between DMOSFET and EMOSFET. 8
5. a) For the given CE amplifier, find the following using r_e model 8
- Input impedance
 - Output impedance
 - Overall voltage gain
- (Assume $\beta = 100$, $r_o = 200 \text{ k}\Omega$)



- b) What is the multistage amplifier circuit? Derive the expression of the total voltage gain of the n -stage cascading of transistors. Why is the CE configuration used in intermediate stages while cascading? 7
6. a) What do you mean by power amplifier? Explain the operation principle and derive efficiency of Class B amplifier with the help of circuit diagram. 8
- b) What are the characteristics of an ideal operational amplifier? Design an operational amplifier to produce the output voltage. 7
- $$V_o = 8V_1 - 10V_2 + 0.5V_3 - 6V_4$$
7. Write short notes on: (Any two) 2×5
- Concept of bias stabilization
 - Emitter follower
 - Clipper and clamping circuit